Deep Levels Characterization in GaN HEMTs—Part II: Experimental and Numerical Evaluation of Self-Heating Effects on the Extraction of Traps Activation Energy

Alessandro Chini, Fabio Soci, Matteo Meneghini, *Senior Member, IEEE*, Gaudenzio Meneghesso, *Fellow, IEEE*, and Enrico Zanoni, *Fellow, IEEE*

***Abstract*— In this paper, the effects of device self-heating on the extraction of traps activation energy are investigated through experimental measurements and numerical simulations. Neglecting the device temperature increase during the experimental measurements can lead to an underestimation of traps activation energies as well as nonoverlapping Arrhenius plots. It will be shown that said artifacts can be removed once device thermal resistance is known and used to correct the temperatures data points at which trap time constants are extracted. The correctness of the proposed method is also supported through numerical simulations carried out both by neglecting and considering thermal effects during the drain current transient measurements. Finally, the experimental results obtained are also suggesting a novel method for the extraction of device thermal resistance, which yielded comparable results with respect to those obtained with other experimental techniques.**

***Index Terms*— Charge carrier processes, gallium nitride, MODFETs, power semiconductor devices, reliability.**

I. Introduction

ALGAN/GAN high electron mobility transistors (HEMTs) outstanding electrical performances are largely surpassing those of other semiconductor-based devices when considering high-power and high-frequency applications [1]. Nevertheless, several research efforts are still to be devoted to further improve both their power capabilities and long-term stability [2]. One of the main issue in limiting GaN HEMT reliability is still represented by the performance losses related to the presence of trapping phenomena due to their detrimental effects on devices large signal characteristics and long-term reliability [2]–[4].

Manuscript received April 9, 2013; revised June 19, 2013 and July 31, 2013; accepted August 6, 2013. Date of publication September 5, 2013; date of current version September 18, 2013. This work was supported in part by the University of Padova under Project ONR N000141010608 "Failure Mechanisms of Gallium Nitride High-Electron Mobility Transistors: A Physics-Based, Nonconventional Approach" under the supervision of Dr. P. Maki, in part by the EDA project MANGA and in part by the Italian MoD project GARANTE. The review of this paper was arranged by Editor T. Palacios.

A. Chini and F. Soci are with the Department of Engineering "Enzo Ferrari," University of Modena and Reggio Emilia, Modena 41125, Italy, and also with the IUNET, University of Bologna, Bologna 40126, Italy (e-mail: alessandro.chini@unimore.it; fabio.soci@unimore.it).

M. Meneghini, G. Meneghesso, and E. Zanoni are with the Department of Information Engineering, University of Padova, Padova 35131, Italy, and also with the IUNET, University of Bologna, Bologna 40126, Italy (e-mail: menego@dei.unipd.it; gauss@dei.unipd.it; zanoni@dei.unipd.it).

Digital Object Identifier 10.1109/TED.2013.2278290

Lattice defects or impurity states within the device structure behave as charge trapping centre, which directly affect the HEMT 2-D electron gas (2DEG) density. They are uniquely identified by charge state, activation, or ionization energy (E_A) and a capture cross section (σ_A). Among these characteristics E_A is considered the signatures of the defect involved in reducing device performance, and its value can typically be deduced by analyzing the emission rate of the trapped charge from the defects, since the E_A of a localized level within the bandgap is related to the rate of thermally activated ionization, by an Arrhenius-type relationship [5]. The correct knowledge of E_A represents a powerful tool for the traps characterization as its value can be uniquely associated with a defects or impurities in the semiconductor. Therefore, the activation energy extraction is a fundamental step toward the defects or impurities identification and the subsequent development of a reliable semiconductor technology.

One of the most commonly used methods for the characterization and identification of traps signature within a field-effect-transistor-based structure is represented by the drain IDLTS测试原理介绍
current transient analysis techniques [6], [7]. Basically, this measurement is carried out by applying voltage pulses to the device terminals to investigate the drain current evolution when the device is switched between two different bias points. Particularly, the device is typically switched from an OFF-state to ON-state condition and the drain current transient is then recorded in time. The aim of this measurement is the analysis of the trap response due to the different steady-state occupations that occurs during the OFF-state and the ON-state bias-points. Particularly, the excess or lack of trapped charges between the two bias points will give rise to a variation in traps occupation during the transition between the two bias points. Said variation will then translate into a drain current variation during time whose time constants will be related to the capture and emission processes of the involved trap levels. Particularly, a transient due to an emission process will have a time constant, which is related to the activation energy of the trap level whose occupancy change is causing the observed variation in drain current. The analysis of drain current transients at different operating temperatures allows to extract the time-constant values associated with the transient (τ) at different temperatures. The experimentally obtained (τ, T) data set allows the construction of an Arrhenius plot,

which can be used for the extraction of the traps activation energy. Since the experimental data used for the Arrhenius plot requires a good knowledge of the device temperature during the drain current transient, it is straight-forward to conclude that any error related to the estimation of the device temperature during the measurement will translate in an error with the extraction of the trap activation energy.

The aim of this paper is thus to exploit the effects of device self-heating during the experimental drain-current transient measurements to highlight the significant differences in the Arrhenius plots that can be obtained neglecting or considering device self-heating effects. It will be shown that Arrhenius plot obtained from different ON-state bias-points, (i.e., different device power levels), yielded different results in terms of activation energy and nonoverlapping (τ, T) data sets. Introducing a temperature values correction based on device self-heating evaluation allowed to remove the artifact in the Arrhenius data plot yielding consistent and over-lapping data sets even when comparing those obtained at different ON-state bias points.

Electro- and electrothermal 2-D numerical simulations were also carried out to support the experimentally observed evidence of the influence of device self-heating on the characterization of trap levels based on the construction of Arrhenius plots. Finally, it will be also shown that the method proposed can also be potentially used for the estimation of device thermal resistance, which can be considered as an alternative to other previously proposed methods [8].

II. Device Structure and Characterization

Devices used in this paper have been fabricated on three different epilayer structures, namely type-A, -B, and -C. All tested devices are single heterojunction AlGaN/GaN HEMTs with a 2 × 50 μm total gate periphery. Type-A and -B structures were grown by MOCVD on a SiC substrate, while type-C was grown on a high-resistivity silicon (111) substrate. After a silicon nitride (SiN) deposition, the device gate foot has been fabricated by Ni/Au metal stack evaporation through SiN passivation in type-A and -B devices, while a Ni/Pt/Au stack has been used in device C. First double-pulse I–V measurements were carried out to extract devices thermal resistances R_{TH} following the experimental method presented in [8]. Thermal resistance values of 85, 59, and 90 K/W were extracted, respectively, for type-A, -B, and -C structures. Note that type-A devices, although being grown on SiC-substrates. yielded a significantly higher R_{TH} value with respect to that of type-B one. This discrepancy might be associated with the different nucleation layer adopted during the material growth, which has been proven to be crucial to reduce device thermal resistance [9].

The drain current transient method used for the traps E_A extraction is similar to the one presented in [6] and it will be briefly described in the following. Initially, the device is biased in ON-state condition through a 100 Ω resistor connected to the drain at $V_{GS} = V_{GH}$ and $V_{DS} = 5$ V. At this point, the gate voltage is pulsed to a negative value V_{GL} to turn off the device. After being turned off for 1 s, the device is again switched on by applying the voltage V_{GH} to the gate terminal, and the

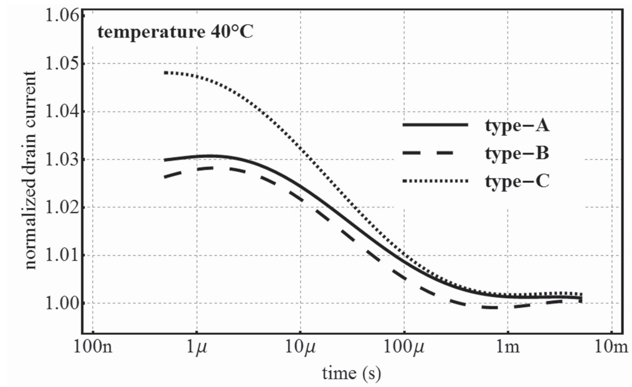

Fig. 1. Drain current transient obtained by pulsing from $V_{GS} = 0$ V, $V_{DS} = 0$ V to $V_{GS} = 0$ V, and $V_{DS} = 5$ V. Transient are showing time constants in the 10–100 μs range, which can be ascribed to the device self-heating effects.

drain current transient is recorded through a digital-sampling-oscilloscope. During all the measurements, the device is placed on a hot-plate to fix the substrate backside temperature.

Drain current transient measurements from the ($V_{GS} = 0$ V, $V_{DS} = 0$ V) quiescent bias point were also performed to evaluate the drain current transient related to device self-heating effects occurring when the device is switched to ON-state condition. Results depicted in Fig. 1 are showing that the drain current level is decreasing with time constants in the 10–100 μs range on all the three tested structures. This suggest that the analysis of trapping phenomena occurring with time constant larger than 100 μs can be considered not affected by thermal transitory effects. On the other hand, trap levels with time constants lower than 100 μs will yield drain current transients affected both by the trapping/detrapping process as well as by the device self-heating effects. In this scenario, it might not be possible to correctly extract the traps-related time constant. A possible solution to this scenario can be obtained by carrying out drain current transient measurement by lowering the base-plate temperature. This will increase the traps time constant and, if a lower enough temperature is applied, move the drain current transient caused by the trap emission/capture process to time constant larger than those involved in the thermal transitory effects. 降低基准温度线不靠谱

Experimental drain current transient obtained for a type-A device with $V_{GH} = 0$ V, $V_{GL} = -10$ V and at a base-plate temperature of 40 °C is depicted in Fig. 2. Two different emission process associated to two different traps can be observed, and will be referred in the following as t_{1A} and t_{2A}. For these two emission process, time constants values of $\tau_{1A} = 1.34$ ms and $\tau_{1B} = 2.52$ s have been measured with a base-plate temperature settled to 40 °C. Fig. 2 also shows the instantaneous dissipated power experimentally measured, which is obtained by multiplying the drain current and drain-source voltage. By knowing the dissipated power and the thermal resistance, it is thus straightforward to estimate the mean channel temperature (T_C) during the measurement as $T_C = T_P + P_D \times R_{TH}$, where T_P is the externally settled plate temperature and P_D the device dissipated power. At this point, some additional comments are needed concerning the

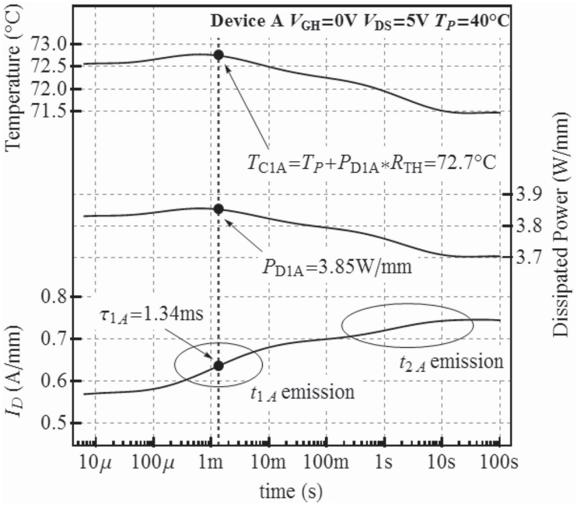

Fig. 2. Measured drain current transient for a type-A device after −10 V negative pulse. Dissipated device power and the estimated mean channel temperature trends are also reported.

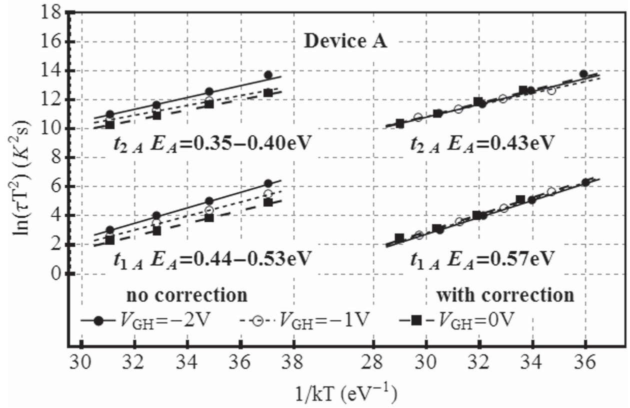

Fig. 3. Device A Arrhenius plots obtained at various V_{GS} values, when applying a gate negative pulse of −10 V. Left: nonoverlapping Arrhenius plots with significantly different ionization energies are obtained at different V_{GH} biases. Right: Arrhenius plots after self-heating correction are showing a better overlap even if data at different V_{GH} levels are compared.

图一中静态偏置点不引入虚栅，所产生的瞬态可认为均是由于热量积累造成的，但图2中引入虚栅，瞬态是由虚栅和耗散功率共同影响的。如何区分？

calculation of the T_C channel temperature during the measured drain current transient. As reported in Fig. 1, type-A device is characterized by an approximately 60 μs time constant for the thermal transient phenomena. We can thus assume that this would be approximately the delay at which the device temperature changes after a change in dissipated power level. In Fig. 4, during the calculation of T_C, we assumed that device temperature is varying instantaneously with respect to the dissipated power. Although this is not correct when the dissipated power is varying significantly over a small time interval, i.e., when the device is turned on, this assumption is somehow reasonable when evaluating the trap-emission time constant, which is causing a dissipated power variation with time constant in the milliseconds range, i.e., more than one decade slower than the thermal transient affecting the device.

As can be seen in Fig. 2, device self-heating induces approximately a 30 °C mean channel temperature increase for type-A device with respect to the settled base-plate temperature. Thus, it is straightforward to notice that neglecting the temperature increase due to device self-heating could lead to a large temperature error especially when studying thermal activated process such as trap emission.

To better evaluate the role of device self-heating on the extraction of activation energy, drain current transient measurements have been performed at different bias points (different V_{GH}). Three different V_{GH} values, 0, −1, and −2 V have been used, which correspond, respectively, to mean power levels of 3.8, 2.5, and 1.1 W/mm, for type-A device. Usually in the extraction of activation energy, self-heating is neglected and the device temperature is assumed equal to the plate temperature, then the Arrhenius plot is drawn using the points (T_P, τ) where τ is the emission process time constant measured for the considered trap at the plate temperature T_P. Using this procedure, the Arrhenius plots depicted in the left part of Fig. 3 (no correction) have been obtained. Three different ionization energies have been found for each trap, with values in the 0.44–0.53 eV range for trap t_{1A} and in the 0.38–0.45 eV range for t_{2A}. Furthermore, the Arrhenius plots obtained at different V_{GH} did not overlap. Because each trap should correspond to a unique Arrhenius plot and a unique activation energy, one might suggest that E_A extraction could be affected by some measurement error. It will be shown in the next section that simply by considering the device self-heating effects, it has been possible to correct the apparent discrepancies observed in the Arrhenius plots and in the extracted ionization energies.

III. Self-Heating Effect Correction

As shown in Fig. 2, type-A device temperature estimated using the mean channel resistance largely differ from the plate temperature. Since charge emission rate of a defect is thermally activated, to obtain the correct value of E_A the device self-heating must be considered. In type-A device, starting from the drain current transient, it is thus possible to estimate the device dissipated power P_{D1A} in correspondence to the time constant τ_{1A} associated with the t_{1A} emission process, as shown in Fig. 2. Assuming a time-independent R_{TH} value, which is 85 K/W for type-A, we can estimate the device temperature T_{C1A} at the time where τ_{1A} is extracted as $T_{C1A} = T_P + P_{D1A} \times R_{TH}$. Although the time independence assumption of R_{TH} is not strictly correct, we can, however, observe that the time constants associated with the emission process are in the order of milliseconds, which are larger than the time constants associated with the self-heating effects (see Fig. 1). Thus, we can speculate that for the device presented in this work and for the emission processes investigated, R_{TH} can be assumed to be reasonably constant. At this point, a new Arrhenius plot was extracted by considering the temperature corrected (T_{C1A}, τ_{1A}) data points instead of the (T_P, τ_{1A}) ones. This correction was also applied for the data referring to the t_{2A} process and for the different V_{GH} measurements levels. As can be seen in the right part of Fig. 3, self-heating corrected data points are now yielding nicely overlapping Arrhenius plots and no significant differences can be observed among the data set obtained at different bias. Moreover, the

本文假定热阻不随时间变化，认为陷阱发射时常数在毫秒量级，大于自热效应时常数

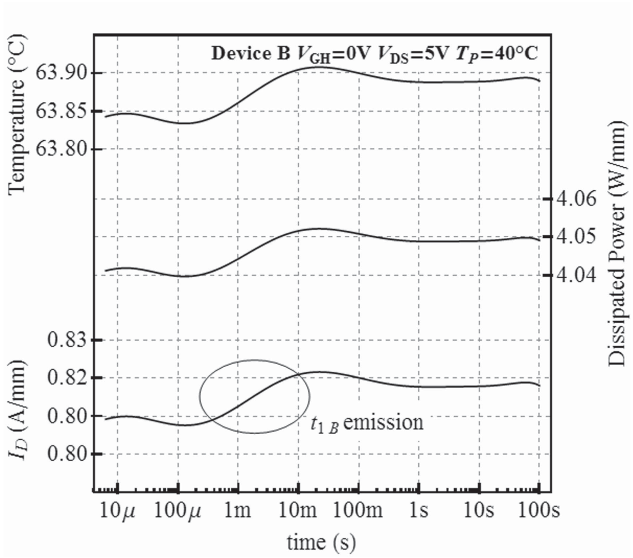

Fig. 4. Device type-B drain current turn-on transients, dissipated power, and estimated channel temperature, after $V_{GL} = -10$ V gate pulse. In this case, drain current transient exhibits only an emission process.

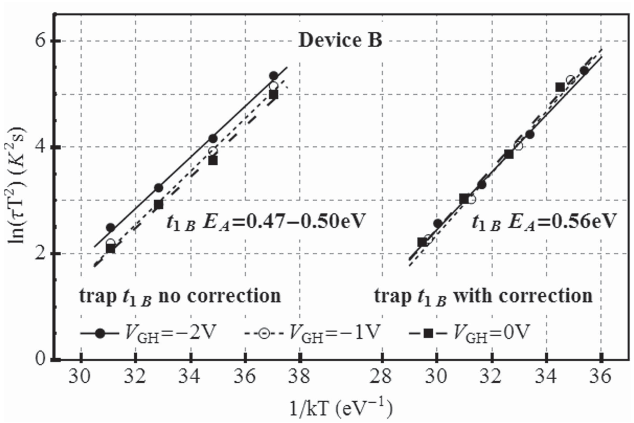

Fig. 5. Device B Arrhenius plots obtained at different V_{GH} values. Arrhenius plots after self-heating correction are showing a better overlap even if data at different V_{GH} levels are compared.

self-heating corrected E_A resulted to be higher than those extracted neglecting the device thermal effects. This increase in the extracted activation energies can simply be explained that when considering the device self-heating the measured emission rate $(1/\tau)$ is not changed but the temperature is increased. Therefore, after the self-heating correction, since the emission rate is the same but the temperature is larger, if we consider the expression of the trap's emission rate reported in [5], it is straightforward that the extracted activation energy must be larger. Traps capture cross-sectional values were also severely affected by device self-heating. Neglecting thermal effects, capture cross-sectional values were in the 1.0×10^{-15} to 3.5×10^{-16} cm^{-2} range for t_{1A} and in the 1.5×10^{-21} to 8.0×10^{-21} cm^{-2} range for t_{2A}. On the other hand, extracted capture cross section after thermal effect correction resulted to be 7×10^{-12} cm^{-2} and 5×10^{-18} cm^{-2} for t_{1A} and t_{2A}, respectively.

The same experimental measurement sets have also been performed on type-B devices. Typical drain current transient

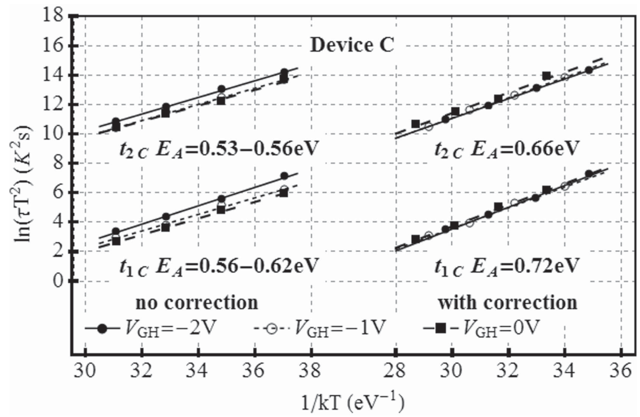

Fig. 6. Device C Arrhenius plots obtained at various V_{GH} values. After self-heating correction Arrhenius plots extracted at different power levels are showing a better overlap, and the extracted ionization energies increase.

for these device are reported in Fig. 4. As we can see only one trap contribution is present: an emission process with time constants in the orders of tens of millisecond. Since the emission time constant is almost three orders of magnitude higher than the time constants associated with the self-heating effects, we can neglect thermal transitory effects. Again a nonoverlapping in the Arrhenius plots and a discrepancy in the extracted values occurs when the device self heating is neglected (see Fig. 5, left side). On the other hand, if we consider the device temperature equal to the sum of the plate temperature plus the self-heating contribution, we can obtain a qualitatively better overlapping of the Arrhenius plots and an increase in the extracted activation energy. For type B device, neglecting thermal effects, capture cross-sectional values were in the 1.0×10^{-15}-5.5×10^{-16}cm^{-2} range for t_{1B}. On the other hand, extracted capture cross section after thermal effect correction resulted to be 6×10^{-12}cm^{-2} for t_{1B}.

Concerning type-C devices a detailed analysis presented in [10] has shown that two different traps are present in these devices. Again, applying the presented self-heating effects correction leaded to a better overlap in Arrhenius plot and to an increase of the measured activation energies as shown in Fig. 6. Similarly to what previously observed, capture cross sections obtained neglecting self-heating effects were in the 3.0×10^{-15} to 1.5×10^{-14} cm^{-2} range for t_{1C} and in the 3.5×10^{-15} to 1.5×10^{-14} cm^{-2} range for t_{2A}. On the other hand, extracted capture cross section after thermal effect correction resulted to be 6×10^{-10} cm^{-2} and 1×10^{-9} cm^{-2} for t_{1A} and t_{2A}, respectively.

IV. Numerical Simulations

Electro- and electrothermal numerical simulations were also carried out by means of the commercially available software DESSIS-ISE. The aim of said simulations is limited to the support and validation of the experimentally observed results concerning the effects of device self-heating on the traps activation energy extraction. For said purpose, a simple GaN HEMT structure including the presence of a surface acceptor trap was simulated. The structure is composed as follows: a 22-nm AlGaN barrier with a 25% Al-concentration; a

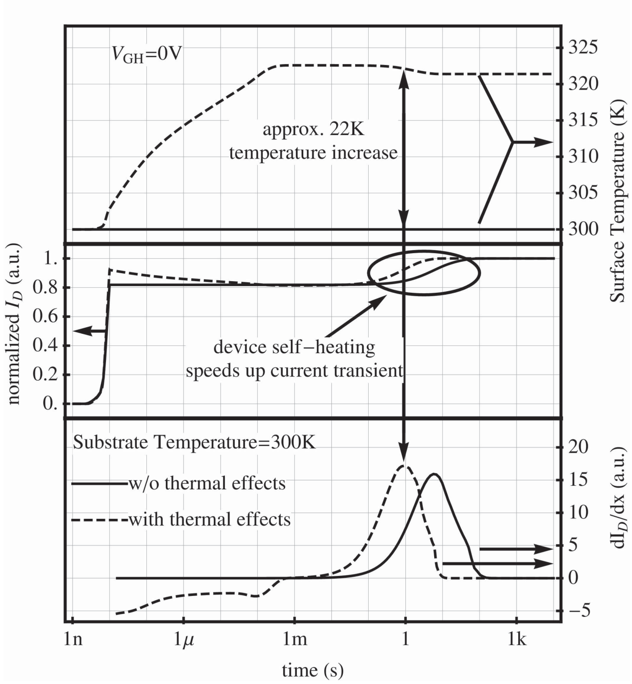

Fig. 7. Electro- [without thermal effects (solid lines)] and electrothermal [with thermal effects (dashed lines)] simulated drain current transient when pulsing the gate terminal from V_{GL}=−10 V to V_{GH} = 0 V at a base-plate temperature of 300 K. The evolution of the average surface temperature and of the dI_D/dt signal are also reported.

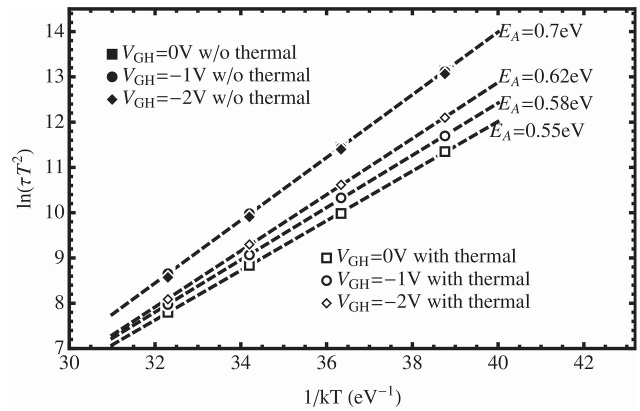

Fig. 8. Arrhenius plots obtained at different V_{GH} values with electro- [without thermal (closed symbols)] and electrothermal [with thermal (open symbols)] numerical simulations. When thermal effects are considered, nonoverlapping Arrhenius plots and lower activation energies are obtained.

1.8-μm GaN channel/buffer layer and a 200-μm-thick SiC substrate, which is connected on its backside to an ideal thermal contact. An acceptor traps with an energy of 0.7 eV from the AlGaN conduction band was included on the device surface to induce the presence of drain current transients when pulsing the device terminals from OFF-state to ON-state conditions [11], [12].

Semiconductor thermal parameters were adjusted according to the data presented in [13]. The dependence of carrier transport parameters was also adjusted to fit the results presented in [14]. It should be stressed, however, that the aim of these simulations is not to quantitatively reproduce the experimental results obtained but rather to further support the strong dependence of the extracted traps activation energy from self-heating effects when a drain-current transient technique is adopted.

Drain current transients were thus simulated by reproducing the same conditions used for the experimental - measurements. A 100-μm gate periphery device was simulated by pulsing the gate terminal from $V_{GL} = -10$ V to different V_{GH} values while biasing the device with a constant voltage source connected to the drain trough a 100 Ω resistor. The level of the constant voltage source was settled to reach a 5-V drain voltage bias when equilibrium is reached (i.e., steady-state conditions).

Results obtained when substrate backside temperature is fixed at 300 K and a V_{GH} level of 0 V are applied to the device are depicted in Fig. 7. Particularly, solid lines represent the data concerning the electrical simulations results while dashed lines refer to the electrothermal ones. Electrical simulations are yielding a drain current transient, which presents a clear transition with time constant in the 10 s range, as confirmed by the calculated dI_D/dt signal peak. On the other hand, electrothermal simulations are first showing a decrease in the drain current level due to device self-heating occurring after that the device is turned on. Note that the temperature transient end approximately at 200 μs from the device turn-on, which is in a good qualitative agreement with both the experimental data presented in this paper and those reported in [15]. Once the device has reached thermal equilibrium, a current transition is observed with time constant in the 1 s range, as confirmed by the calculated dI_D/dt signal peak. The effect of device self-heating is thus to speed ups the traps emission process due to the higher average surface temperature, approximately a 22 K increase, when compared with the simulation obtained neglecting thermal effects.

Simulations were then carried out at different substrate backside temperatures and different V_{GH} levels, while time constants were then extracted by analyzing the dI_D/dt signal peak. The resulting Arrhenius plots for the different simulated conditions are summarized in Fig. 8. Particularly, it can be seen that electrical simulations are yielding a nice overlapping Arrhenius plot (closed symbols) regardless of the V_{GH} levels used for trap characterization. Moreover, the 0.7 eV extracted activation energy corresponds to the trap energy level, which as inserted in the AlGaN device surface. On the other hand, Arrhenius plots at different V_{GH} obtained with electrothermal simulations are not overlapping, and the extracted activation energies are lower than the 0.7 eV value. Simulations results are thus yielding the same qualitative trend observed during the experimental measurements previously presented: device self-heating can significantly change the traps Arrhenius plots when measured at different ON-state power levels. To further proof the dependence of the Arrhenius plot from device self-heating, electrothermal data were adjusted by considering the simulated average surface temperature at the time corresponding to the dI_D/dt peak rather than the substrate backside temperature. The adoption of this correction shifts the data points obtained with electrothermal simulations to higher temperature and they all line up with the Arrhenius

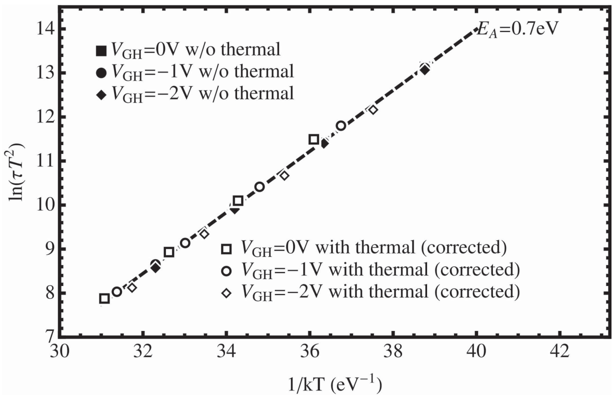

Fig. 9. Arrhenius plots obtained at different V_{GH} values with electro- [without thermal (closed symbols)] and electrothermal [with thermal (corrected)-open symbols] numerical simulations. Considering the average surface temperature instead of the base-plate temperature allows to obtain overlapping Arrhenius plots and correct activation energy value.

plot obtained with electrical-only simulations (see Fig. 9). This result further confirms that the apparent discrepancy of the Arrhenius plot extracted at different ON-state bias-points is solely related to the erroneous evaluation of the temperature at which the trap time constant are extracted.

V. Novel Method for Device Thermal Resistance Estimation

Experimental results are also suggesting a novel method for the extraction of device thermal resistance (R_{TH}). According to the experimental and numerical results that have been presented we have observed that the R_{TH} value, influences the position of the Arrhenius plots in the ($\ln(\tau T^2)$, $1/kT$) plane when self-heating correction is applied. Let us consider now the first trap t_{1A} of type-A device. Neglecting self-heating effects (i.e., assuming $R_{\mathrm{TH}} = 0$ K/W) yields non overlapping Arrhenius plots when different ON-state bias points are considered. On the other hand, if R_{TH} is chosen equal to the experimentally measured vale (85 K/W for type-A devices) a reasonably good qualitative overlap is obtained. Furthermore, a larger value of R_{TH} is used for self-heating correction, i.e., 180 K/W; the Arrhenius plots will again not overlap as shown in Fig 10. With the assumption that to each trap level should corresponds a unique Arrhenius plot, we can thus speculate that the correct value of R_{TH} could be estimated by evaluating the alignment of the Arrhenius plots measured at different power levels.

Linear regression has thus been applied to the set of points (T_{C1A}, τ_{1A}) measured at the different power levels for the first emission process of device A. Then the coefficient of determination, R^2, of the linear least-square regression has been used to quantitatively analyze the point alignment. As shown in Fig. 11(a), the R^2 versus R_{TH} graph for the t_{1A} traps shows a bell-shape, with a maximum at 80 K/W, solid lines. Instead, if we consider the data sets related to the t_{2A} trap, the best alignment is obtained for $R_{\mathrm{TH}} = 87$ K/W; see Fig. 11(a) dashed lines. Therefore, considering the R^2 parameter a channel resistance of 80 and 87 K/W can be estimated considering, respectively, trap t_{1A} and t_{2A}. Note that these two

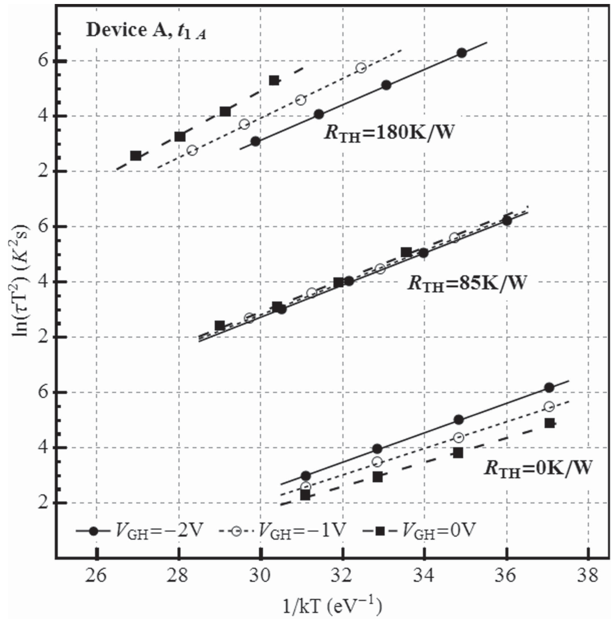

Fig. 10. Self-heating corrected Arrhenius plots obtained for the trap t_{1A} of type-A device at different R_{TH} values. When R_{TH} is chosen higher or lower than the measured value (85 K/W), the Arrhenius plots do not overlap.

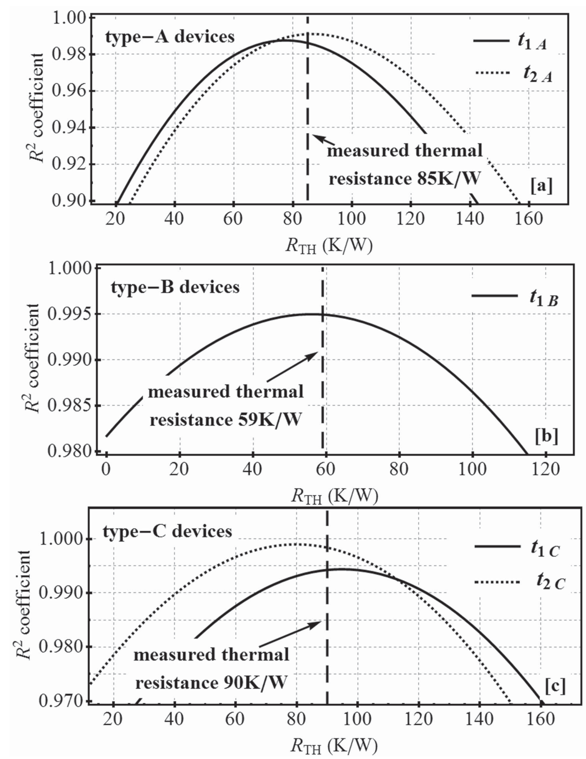

Fig. 11. R^2 dependency on R_{TH} for the different type of devices tested. R^2 refers to the least square linear regression of the Arrhenius points obtained at $V_{\mathrm{GH}}-2$, -1, and 0 V.

values extracted by analyzing the alignment of the Arrhenius plots are quite closed to the one experimentally measured (85 K/W) with the technique presented in [8].

A similar analysis has been carried out also for type-B and type-C devices. Particularly, as can be seen in Fig. 11(b), a thermal resistance of 56 K/W has been obtained, which again is quite close to the 59 K/W value extracted with the technique presented in [6]. Concerning type-C devices, values of 93 and 83 K/W have been obtained by analyzing trap t_{1C} and t_{2C}, respectively; see Fig. 11(c), which are again quite close to the 90 K/W value extracted with the technique presented in [8].

We can thus state that, at least for the device presented in this paper, the maximum value of the coefficient of determination R^2, which corresponds to the best Arrhenius plots overlap, is obtained for a value of R_{TH} very close to those experimentally measured with the method presented in [8]. The analysis of the R^2 coefficient can thus be used to experimentally estimate the device thermal resistance as an alternative method to [8], once at least two or more Arrhenius plots measured at different powers level are available.

It should be stressed, however, that the technique proposed in this paper might be subjected to systematical errors due to the location of the trap levels whose emission time constants are used for the construction of the Arrhenius plot and the subsequent R_{TH} extraction based on the R_2 coefficient. Because of the temperature variation across the device structure, i.e., from source to drain and from surface to GaN/SiC interface, trap levels located close to the drain-edge of the gate contact will be subjected to slightly higher temperatures with respect to those located far away from this region, which roughly corresponds to the hottest part of the device. As a consequence, R_{TH} values extracted with the proposed method might be overestimated or underestimated if the emission time constants refers to a trap located closed to the device hot-point or far away from it, respectively. Different traps spatial location might be the cause of the different R_{TH} values obtained when using different traps for the extraction of device thermal resistance [see Fig. 11(a) and (c)].

VI. Conclusion

In conclusion, an experimental and numerical analysis of the relation between device self-heating and the extraction of traps activation energies using the drain current transient technique has been presented. It has been shown that device self-heating can induce the extraction of lower activation energies values with respect to the correct ones as well as nonoverlapping Arrhenius plot when different ON-state bias point are used. Numerical simulations confirmed all the speculations made during the discussion of the experimentally obtained results. Moreover, a novel technique for the estimation of device thermal resistance based on the analysis of Arrhenius plot overlapping has also been presented and discussed. Experimental results obtained on three different wafers applying said novel techniques are yielding comparable results with respect to those obtained through previously proposed thermal resistance characterization techniques [8].

Acknowledgment

The authors would like to thank A. Cetronio, C. Lanzieri, A. Pantellini, A. Nanni, and M. Peroni (Selex-ES, Italy) for providing test structures and for useful discussions.

References

[1] U. K. Mishra, L. Shen, T. E. Kazior, and Y.-F. Wu, "GaN-based RF power devices and amplifiers," *Proc. IEEE*, vol. 96, no. 2, pp. 287–305, Feb. 2008.

[2] G. Meneghesso, M. Meneghini, A. Tazzoli, N. Ronchi, A. Stocco, A. Chini, and E. Zanoni, "Reliability issues of gallium nitride high electron mobility transistors," *Int. J. Microw. Wireless Technol.*, vol. 2, no. 1, pp. 39–50, 2010.

[3] I. Daumiller, D. Theron, C. Gaquière, A. Vescan, R. Dietrich, A. Wieszt, H. Leier, R. Ventury, U. K. Mishra, I. P. Smorchkova, S. Keller, N. X. Nguyen, C. Nguyen, and E. Kohn, "Current instabilities in GaN-based devices," *IEEE Electron Device Lett.*, vol. 22, no. 2, pp. 62–64, Feb. 2001.

[4] R. Vetury, N. Q. Zhang, S. Keller, and U. K. Mishra, "The impact of surface states on the dc and RF characteristics of AlGaN-GaN HFETs," *IEEE Trans. Electron Devices*, vol. 48, no. 3, pp. 560–566, Mar. 2001.

[5] D. V. Lang, "Deep-level transient spectroscopy: A new method to characterize traps in semiconductors," *J. Appl. Phys.*, vol. 45, no. 7, pp. 3023–3032, Jul. 1974.

[6] M. Ťapajna, R. J. T. Simms, Y. Pei, U. K. Mishra, and M. Kuball, "Integrated optical and electrical analysis: Identifying location and properties of traps in AlGaN/GaN HEMTs during electrical stress," *IEEE Electron Device Lett.*, vol. 31, no. 7, pp. 662–664, Jul. 2010.

[7] J. Joh and J. A. del Alamo, "A current-transient methodology for trap analysis for GaN high electron mobility transistors," *IEEE Trans. Electron Devices*, vol. 58, no. 1, pp. 132–140, Jan. 2011.

[8] J. Joh, J. A. del Alamo, U. Chowdhury, T.-M. Chou, H.-Q. Tserng, and J. L. Jimenez, "Measurement of channel temperature in GaN high-electron mobility transistors," *IEEE Trans. Electron Devices*, vol. 56, no. 12, pp. 2895–2901, Dec. 2009.

[9] A. Manoi, J. W. Pomeroy, N. Killat, and M. Kuball, "Benchmarking of thermal boundary resistance in AlGaN/GaN HEMTs on SiC substrates: Implications of the nucleation layer microstructure," *IEEE Electron Device Lett.*, vol. 31, no. 12, pp. 1395–1397, Dec. 2010.

[10] A. Chini, F. Soci, F. Fantini, A. Nanni, A. Pantellini, C. Lanzieri, D. Bisi, G. Meneghesso, and E. Zanoni, "Field plate related reliability improvements in GaN-on-Si HEMTs," *Microelectron. Rel.*, vol. 52, nos. 9–10, pp. 2153–2158, Sep./Oct. 2012.

[11] A. Chini, V. Di Lecce, M. Esposto, G. Meneghesso, and E. Zanoni, "Evaluation and numerical simulations of GaN HEMTs electrical degradation," *IEEE Electron Device Lett.*, vol. 30, no. 10, pp. 1021–1023, Oct. 2009.

[12] M. Faqir, G. Verzellesi, A. Chini, F. Fantini, F. Danesin, G. Meneghesso, E. Zanoni, and C. Dua, "Mechanisms of RF current collapse in AlGaN–GaN high electron mobility transistors," *IEEE Trans. Device Mater. Rel.*, vol. 8, no. 2, pp. 240–247, Jun. 2008.

[13] J. Das, H. Oprins, H. Ji, A. Sarua, W. Ruythooren, J. Derluyn, M. Kuball, M. Germain, and G. Borghs, "Improved thermal performance of AlGaN/GaN HEMTs by an optimized flip-chip design," *IEEE Trans. Electron Devices*, vol. 53, no. 11, pp. 696–702, Nov. 2006.

[14] M. Farahmand, C. Garetto, E. Bellotti, K. Brennan, M. Goano, E. Ghillino, G. Ghione, J. Albrecht, and P. Ruden, "Monte Carlo simulation of electron transport in the III-nitride wurtzite phase materials system: Binaries and ternaries," *IEEE Trans. Electron Devices*, vol. 48, no. 3, pp. 535–542, Mar. 2001.

[15] F. Bertoluzza, N. Delmonte, and R. Menozzi, "Three-dimensional finite-element thermal simulation of GaN-based HEMTs," *Microelectron. Rel.*, vol. 49, no. 5, pp. 468–473, 2009.

Authors' photographs and biographies not available at the time of publication.